The Constants of Nature

legend

$t_p = 5.39125836832313\ldots \times 10^{-44}$ s — Planck time
$l_p = 1.61625918175645\ldots \times 10^{-35}$ m — Planck length
$q_p = 1.87554596713962\ldots \times 10^{-18}$ C — Planck charge
$T_p = 1.41678698590795\ldots \times 10^{32}$ K — Planck temperature
$m_p = 2.17642683817579\ldots \times 10^{-8}$ kg — Planck mass

$\boxtimes = 0.999999199973626\ldots \times 10^{-7}$ — hyperbolic inversion boundary
5 — number of dimensions
7 — break in scale symmetry

$ж_1 = 0.0854245431533304\ldots$ — 1st hyperbolic partition constant
$ж_2 = 3.66756753485501\ldots$ — 2nd hyperbolic partition constant
$ж_3 = -1.87649603900417\ldots + 4.06615262615972\ldots i$ — 3rd hpc
$ж_4 = -1.87649603900417\ldots - 4.06615262615972\ldots i$ — 4th hpc
$ж_r = 4.47826244916751\ldots$ — hyperbolic radius constant
$ж_\theta = 2.00316562310924\ldots$ — hyperbolic radian constant

$G_{Gi} = 1.01494160640965\ldots$ — Gieseking's constant
$V_{fe} = 2.02988321281930\ldots$ — figure eight knot hyperbolic volume
$K = 0.915965594177219\ldots$ — Catalan's constant
$C_d = 0.291560904030818\ldots$ — domino tiling constant
$D_d = 1.79162281206959\ldots$ — dimer constant
$\omega_1 = 0.764977018528596\ldots + 1.32497062714087\ldots i$ — omega_1 constant
$\omega_2 = 1.529954037057192\ldots$ — omega_2 constant

$i^{i^{i^{\cdot^{\cdot^{\cdot}}}}} = 0.438282936727032\ldots + 0.360592471871385\ldots i$ — power tower of i
$i = (-1)^{1/2}$ — imaginary unit
$\varphi_i = (-1)^3$ — imaginary golden ratio
$\varphi = 1.61803398874989\ldots$ — golden ratio
$\pi = 3.14159265358979\ldots$ — Archimedes' constant
$P_{up} = 2.29558714939263\ldots$ — universal parabolic constant
$s = 5.24411510858423\ldots$ — arc length of the unit lemniscate
$L = 2.622057554292119\ldots$ — lemniscate constant
$L_1 = 1.31102877714605\ldots$ — 1st lemniscate constant
$L_2 = 0.599070117367796\ldots$ — 2nd lemniscate constant
$C_U = 0.847213084793979\ldots$ — ubiquitous constant
$G_{Ga} = 0.834626841674073\ldots$ — Gauss's constant
$W_{We} = 0.474949379987920\ldots$ — Weierstrass constant

$C_{CF} = 0.697774657964819\ldots$ — continued fraction constant
$C_{R1} = 0.998136044598509\ldots$ — Ramanujan's 1st continued fraction constant
$\mathcal{L}_{Li} = 0.110001000000000\ldots$ — Liouville's constant
$M = -1.74756459463318\ldots$ — Madelung constant
$\mathcal{P} = 1.22674201072035\ldots$ — prime constant

$e = 2.71828182845904\ldots$ — Euler's number
$\gamma = 0.577215664901532\ldots$ — Euler-Mascheroni constant
$S = 0.822825249678847\ldots$ — Sierpiński's constant
$D_{Do} = 0.739085133215160\ldots$ — Dottie number
$L_{LL} = 0.662743419349181\ldots$ — Laplace limit
$C_{CFP} = 1.19967864025773\ldots$ — real fixed point of the hyperbolic cotangent
$j_{0,1} = 2.404825557695772\ldots$ — 1st root of the Bessel function

$\alpha_F = 2.50290787509589\ldots$ — alpha Feigenbaum constant
$\delta_F = 4.66920160910299\ldots$ — delta Feigenbaum constant
$\mathcal{S} = 3.24697960371714\ldots$ — silver constant
$P = 1.32471795724474\ldots$ — plastic constant
$C_Q = 0.605443657196732\ldots$ — QRS constant
$x_\infty = 2.29316628741186\ldots$ — 1st Foias constant
$K_{-6} = 1.15655237442151\ldots$ — Khinchin mean of order −6

$T_0 = 2.73150000000000\ldots \times 10^2$ kelvin — freezing point of water
$p_0 = 1.00000000000000\ldots \times 10^5$ pascal — 1 bar
$p_1 = 1.01325000000000\ldots \times 10^5$ pascal — standard atmospheric pressure
$E_1 = 6.09110229711386\ldots \times 10^{-24}$ joules — energy associated with $\Delta_{\nu_{Cs}}$
$f_1 = 5.40000000000000\ldots \times 10^{14}$ Hz — frequency of max light conversion
$\lambda_{peak} = 4.96511423174427\ldots$ — peak λ of spectral radiance/unit wavelength
$\nu_{peak} = 2.82143937212207\ldots$ — peak emission of spectral flux/unit frequency

$n!$ = factorial function
$!n$ = derangement function
$\log(x)$ = natural logarithm
$\Gamma(x)$ = gamma function
$\zeta(x)$ = Riemann zeta function

lm = lumen = cd · sr
N = newton = m kg/s²
$\mathcal{N}$ = noether = m kg/s
Ω = ohm = m² kg/s C²
Pa = pascal = kg/s² m
S = siemens = s C²/m² kg
T = tesla = kg/s C
V = volt = m² kg/s² C
W = watt = m²kg/s³
Wb = weber = m² kg/s C

units

s = second
m = meter
C = coulomb
K = kelvin
kg = kilogram

composite units

A = ampere = C/s
F = farad = s² C²/m² kg
H = henry = m² kg/C²
Hz = hertz = 1/s
J = joule = m² kg/s²

derived units

MHz = megahertz = 1.00000000000000 … × 10⁶ 1/s
fm = femtometer = 1.00000000000000 … × 10⁻¹⁵ m
eV = electron-volt = 1.00000000000000 … eV
MeV = megaelectron-volt = 1.00000000000000 … × 10⁶ eV
GeV = gigaelectron-volt = 1.00000000000000 … × 10⁹ eV

$$\frac{r_n}{r_e} = \left(\frac{4\pi}{\Gamma(5)}\right)^2 \qquad \frac{r_n}{r_+} = K \qquad \frac{r_n}{r_\alpha} = \sqrt{\frac{C_U}{4}} \qquad \frac{r_e}{r_{de}} = \sqrt{\frac{2\pi}{\log(36)}} \qquad E_p = \frac{l_p^2 m_p}{t_p^2}$$

transform space

$$V_{24} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)^1\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

figure eight knot hyperbolic volume

$$V_{fe} = i\left[Li_2\left(\frac{1}{\varphi_i}\right) - Li_2(\varphi_i)\right]$$

binomial constructor

$$A_{ext}\,B_{ext}\,(1 + A_{int}\,\boxtimes) \qquad \boxtimes = \frac{l_p m_p}{q_p^2} \times \frac{\text{coulomb}^2}{\text{meter kilogram}}$$

hyperbolic partition equation

$$\frac{1}{x} + x + \frac{x^3}{2\pi} = (i^i)^{-\frac{4\pi}{8}} - \frac{m_p}{\text{kilogram}}$$

$ж_1ж_2ж_3ж_4 = 2\pi$
$ж_1 + ж_2 + ж_3 + ж_4 = 0$
$ж_1^2 + ж_2^2 + ж_3^2 + ж_4^2 = -4\pi$

$\alpha = ж_1^2$
$e = ж_1 q_p$

Physics Monastery
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